Assignment # 1

BEM Model

Course	Rotor/Wake Aerodynamics AE4135	
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1 Project Introduction

The aim of this report is to document and analyses the results of the developed Blade Element Momentum (BEM) model that is applied to a **propeller**. This section gives a brief overview of the report and defines the main propeller specifications of the propeller.

Firstly, the structure and hierarchy of the developed BEM model is detailed with the help of a flowchart. Then, after proper code verification with existing and tested BEM codes, the assumptions that were made and their implications are discussed. The model is then used to evaluate the given rotor at different conditions. Furthermore, the radial distribution of stagnation pressure will be analyzed at infinity upstream, infinity downstream, upstream at the blade and downstream at the blade. After which the operational point of the airfoil is analyzed in terms of lift and drag.

Furthermore, an optimization was performed to enhance the energy harvesting capacity of the propeller in landing conditions. Finally, a conclusion highlighting the impact of operational variables and BEM model assumptions on the results is presented.

Rotor Specifications

The cross-sectional shape of a rotor blade is the *ARA-D8%* airfoil and the geometry is specified by a linear function of twist angle and chord length. The hub is located at $(r/R)_{\text{hub}} = 0.25$. The propeller has 6 blades and a radius of 0.7 meters and a collective blade pitch of 46° at $r/R = 0.70$. These specifications are summarized in the following table:

Table 1: Rotor Specifications

Parameter	**Value**
Radius	0.70 m
Number of blades	6
Blade start	$r/R = 0.25$ (hub for $r/R < 0.25$)
Twist	$-50(r/R) + 35°$ (for $r/R > 0.25$)
Collective pitch	46° at $r/R = 0.70$
Chord distribution	$0.18 - 0.06(r/R)$ (for $r/R > 0.25$)
Airfoil	ARA-D8%

Operational Specifications

The operational specifications for the propeller are summarized in the table below:

Table 2: Operational Specifications

Parameter	**Value**
Freestream velocity U_0	60 m/s
Rotational speed	1200 RPM
ISA altitude	2000 m
Freestream incidence angle	0°
Rotor yaw angle	0°

2 BEM Model Flowchart

In this section the flow chart of the BEM model is presented and a general overview is provided. It is important to note that in this report we use the definitions of induction factors a, a' for propellers. Thus, in normal operation the normal and tangential velocity components at the rotor plane are $V_x(1+a) = V_\infty(1+a)$ and $V_y(1-a') = \Omega r(1-a')$.

The BEM model is initialized by defining the twist distribution, the chord distribution, geometrical properties, operating condition, discretization method and the 2 dimensional airfoil polars. The model first discretizes the blade into a specified number of annuli from $r/R = (r/R)_{\text{hub}}$ to $(r/R) = 1$. For each annulus, the code performs and converges a BEM method. The converged output properties include α, ϕ, a, a', C_T, C_x, C_y, C_q, and F. The BEM method was based on the work of Ning [1], where he proposes to converge the inflow angle ϕ instead of both the induction factors a and a' at the same time. This reduces the problem dimensionality of the problem, allowing the use of robust root-finding schemes that have guaranteed convergence, such as a bracketing scheme. This bracket set at the first quadrant $[0, \pi/2]$.

With this method, the residual function $\mathcal{R}(\phi)$ is defined for the root-finding scheme as follows:

$$\tan\phi = \frac{V_x}{V_y}\frac{(1+a)}{(1-a')} \rightarrow \mathcal{R} = \frac{\sin\phi}{(1+a)} - \frac{V_y}{V_x}\frac{\cos\phi}{(1-a')} \tag{1}$$

Given an initial guess of ϕ, α is calculated from the pitch angle, after which $C_l(\alpha)$ and $C_d(\alpha)$ can be determined from the provided polar. Subsequently, these forces are rotated in the normal-tangential reference system C_x and C_y using the inflow angle. Then the hub and tip correction F based on Prandtl [2], is computed. This particular correction is applied *directly to the thrust and torque*, as done by [1, Eq 3, 4]. The non-dimensional parameters κ and κ' are computed based on eqs. (2) and (3), from which the induction factor a and a' are derived, respectively. For the axial induction factor in particular, the Prandtl heavily loaded-turbine correction is applied.

$$\kappa(a) = \frac{C_x\,\sigma}{4F\sin^2\phi} \tag{2}$$

$$\kappa'(a') = \frac{C_y\,\sigma}{4F\sin\phi\cos\phi} \tag{3}$$

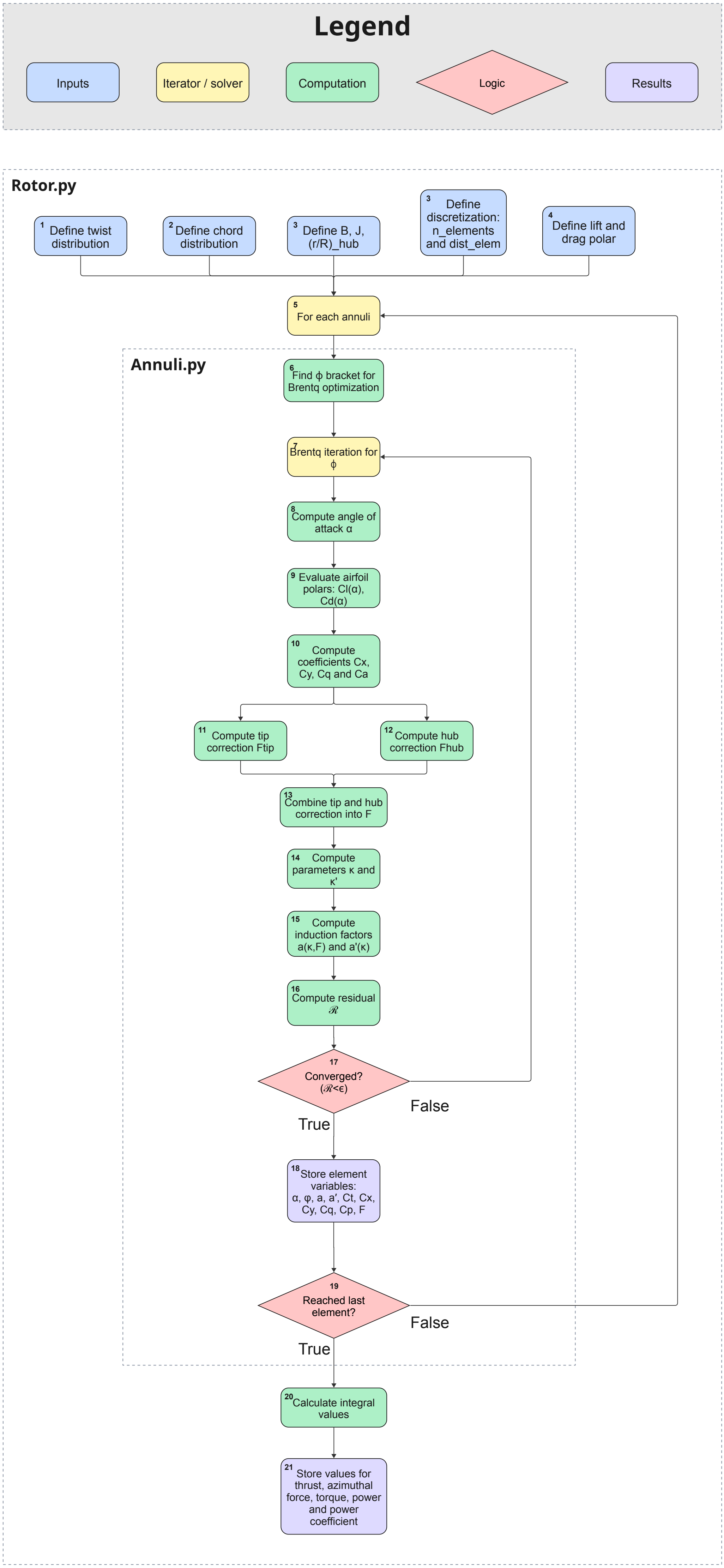
The residual can now be calculated, following eq. (1), which is returned to the root-finding algorithm, which was chosen to be the Bentq bracketing scheme.

Once each annuli has been converged, the total rotor forces and moments are computed by integrating the contributions for all annuli.

3 Code Verification

To verify the correct implementation of the BEM code, the results were verified against two established models:

- BEM code of Ning [1].
- JavaProp.


Legend
Inputs
Iterator / solver
Computation
Logic
Results
Rotor.py
1 Define twist distribution
2 Define chord distribution
3 Define B, J, (r/R)_hub
3 Define discretization: n_elements and dist_elem
4 Define lift and drag polar
5 For each annuli
Annuli.py
6 Find ϕ bracket for Brentq optimization
7 Brentq iteration for ϕ
8 Compute angle of attack α
9 Evaluate airfoil polars: Cl(α), Cd(α)
10 Compute coefficients Cx, Cy, Cq and Ca
11 Compute tip correction Ftip
12 Compute hub correction Fhub
13 Combine tip and hub correction into F
14 Compute parameters κ and κ'
15 Compute induction factors a(κ,F) and a'(κ)
16 Compute residual ℛ
17 Converged? (ℛ<ε)
False
True
18 Store element variables: α, φ, a, a', Ct, Cx, Cy, Cq, Cp, F
19 Reached last element?
False
True
20 Calculate integral values
21 Store values for thrust, azimuthal force, torque, power and power coefficient

3.1 Ning Verification

The spanwise distribution plots for several variables of interest are shown in fig. 1, for the propeller rotor subject to study in this assignment table 1 at default operating conditions table 2($J = 2.14$). As can be seen, the spanwise distribution for both method match every except exactly at the hub and tip. This gives confidence that the code is correctly implemented.

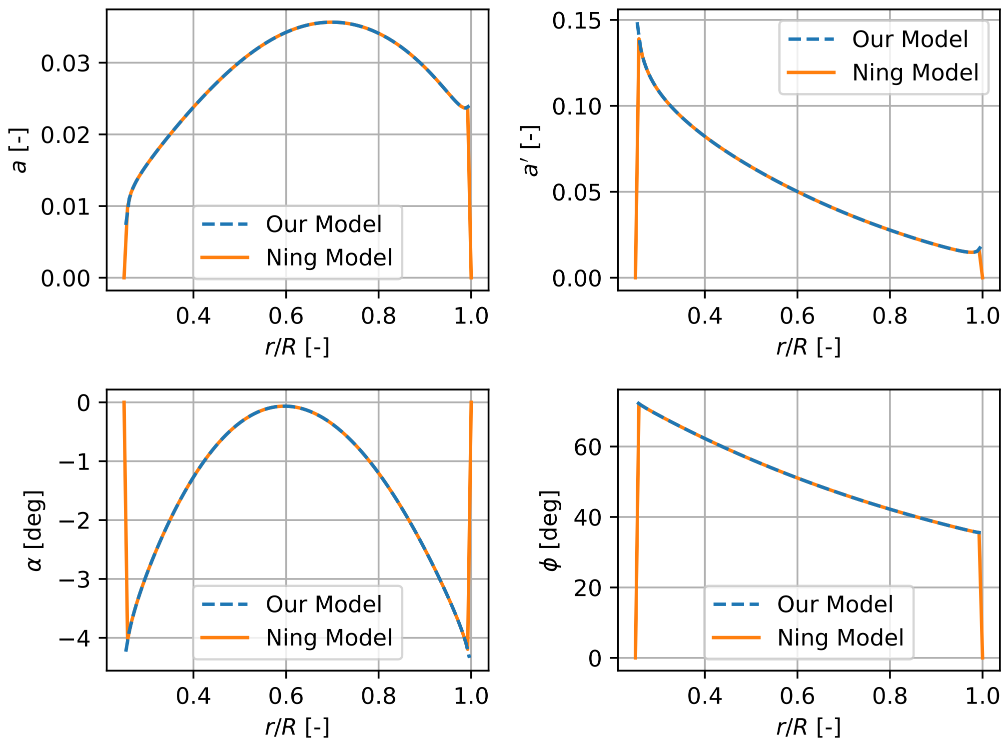


Figure 1: Code verification against Ning

3.2 Javaprop Verification

Furthermore, the code was also verified against JavaProp. For this comparison, a different airfoil was used, more specifically, the ARAD6% instead of the ARAD8%. The polar for the different airfoil was taken from [3]. The same rotor geometry and operational conditions were used in both cases which can be found tables 1 and 2. The resulting plot of total thrust and torque against the advance ratio can be seen below.

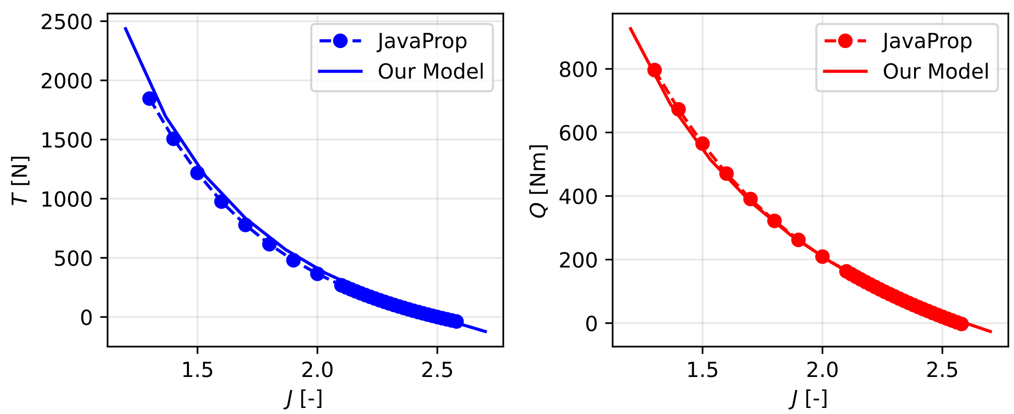


Figure 2: Code verification against JavaProp

From Figure 2 it can be seen that both blade element models output nearly the same results. The small discrepancies can be explained by the Reynolds number differences, as the polar from [3] uses a $Re = 1,000,000$, and JavaProps only offered up to $Re = 100,000$ for the airfoil. In both simulations, the Reynolds number is assumed to be constant along the whole blade. Again, these results give confidence that the developped BEM code has been implemented correctly.

4 Main Assumptions

To create a successful and efficient BEM model, several important assumptions need to be made. These assumptions allow for efficient simulations of the rotor and wake physics while reducing the required computational power. The main assumptions are listed below.

- **Independence of annuli**: No aerodynamic interactions between different sections of the blade are considered. This implies that there is no radial flow, and the force on the blade element is solely responsible for the change of momentum across the annulus. In reality, however, radial pressure gradients and vorticity conservation induce radial flow and coupling between sections, especially near the blade root and tip or under high loading conditions. Neglecting these effects violates the continuity of the wake vortex system and can lead to inaccuracies in the predicted induction and loading distributions. This assumption is only really valid if the axial induction factor does not vary radially [4]. Although this is rarely the case, in practice, the assumption is acceptable for tip-speed ratios smaller than 3 [4]. This decouples the blade sections and allows for each section to be analyzed separately and independently.

- **Forces on the blade element can be calculated from 2D airfoil characteristics** based on angle of attack. This assumption is tied to the independence of the annuli. It implies that 3D rotation and finite aspect ratio effects are not considered, leading to errors near the root and tip. For this, tip and hub corrections are applied that reduce the loading to zero at the tip and root.

- **Inflow assumed uniform and aligned with the rotor axis**. This neglects yawed operation, leading to a a model with a uniform azimuthal distribution.

- **Steady flow**. Dynamic effects are not considered, implying that the results are not representative in dynamic changes of inflow condition and pitch changes.

- **Inviscid flow**. The flow in the stream-tube is assumed inviscid, but viscosity is indirectly accounted for in the airfoil polar. It is important to stress that there is no modeling of the losses associated with wake mixing and energy dissipation in the wake.

- **Incompressible flow**, enabling, with the inviscid assumption, to use Bernoulli and momentum theory. This assumption is valid for low Mach numbers, often recognized for $M < 0.3$. Given the operating conditions in table 2, $V = 60$ m/s and $\Omega R = 87$ m/s, the effective velocities is in the order of 100 m/s. Thus, although we assume incompressible flow, we expect compressibility effects at the tip. Nonetheless, these effects are local and limited.

- **No heat transfer**. The only energy transfer is mechanical. This is acceptable because there are no significant temperature changes and the speed is low.

- **Rigid blades.** The deformation of the blades with the aerodynamic loads is not considered. This deformation would change the angle in attack and hence loading distribution.

5 Results

This section contains all the results obtained using the BEM model that address the posed questions, accompanied by the relevant discussions. This includes:

1. influence of tip corrections
2. spanwise distributions of selected variables of interest for J =1.6, 2.0 and 2.4.
3. total thrust and torque versus the advance ratio J.
4. convergence and discretization matters, including the influence of the number of elements and the spacing method.

5.1 Hub and Tip Corrections

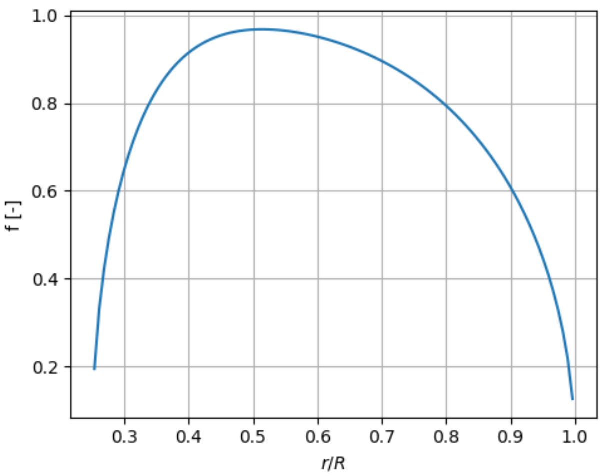
The correction applied in the BEM model is that of Prandtl-Glauert used by Ning [1]. The methods corrects for losses near the hub and the tip:

$$f_{tip} = \frac{B}{2}\left(\frac{R_{tip} - r}{r|\sin\phi|}\right) \qquad f_{hub} = \frac{B}{2}\left(\frac{r - R_{hub}}{R_{hub}|\sin\phi|}\right)$$

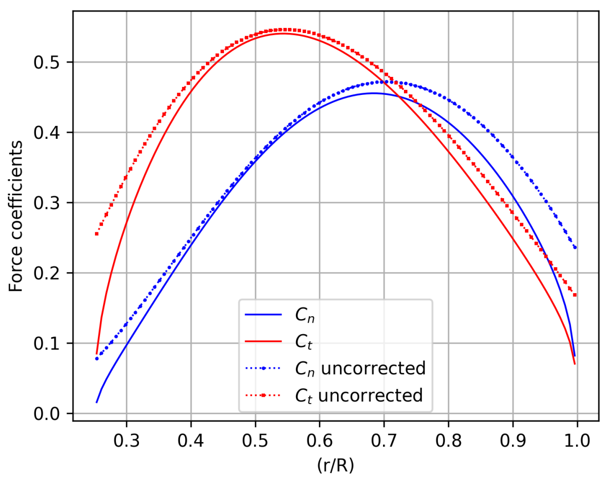
$$F_{tip} = \frac{2}{\pi}\arccos(\exp(-f_{tip})) \qquad F_{hub} = \frac{2}{\pi}\arccos(\exp(-f_{hub}))$$

$$F = F_{tip}F_{hub}$$

The distribution of the correction function can be seen in Figure 3a.



(a) Tip correction function



(b) Influence of tip corrections on normal and tangential force coefficients, letting $F = 1$ in the uncorrected case for $J = 2$.

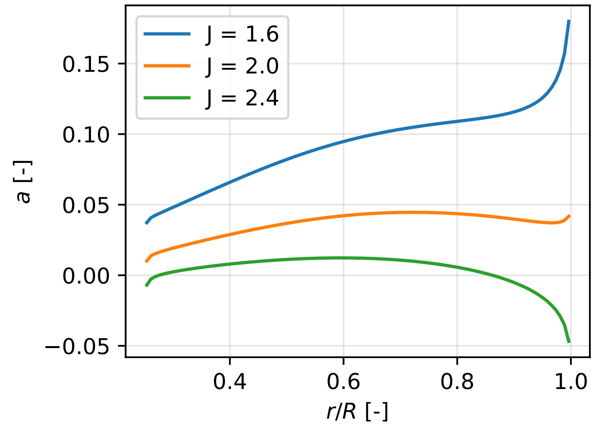
Figure 3: Tip correction factor and its influence on blade loads.

It is important to note that the form of this correction differs from the one given in the slides, as it is applied to the thrust and torque directly instead of the induction factors a and a'. The influence of the tip corrections can be seen in Figure 3b.

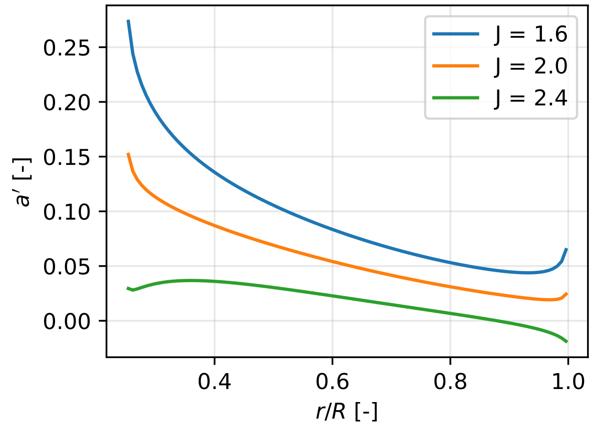
As can be seen, the tip corrections reduce the loading to zero at the hub and tip, representing the effects a finite blade and a shed vortex sheet. Overall, this necessary correction reduces the total thrust and azimuthal forces.

5.2 Spanwise Coefficient Distributions

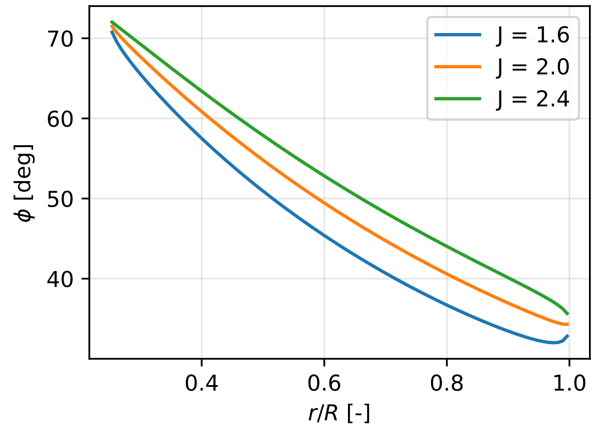
The propeller performance and operating conditions are greatly influenced by the advance ratio. In this assignment $J = 1.6$, 2.0, and 2.4 will be considered. The variables of interest can be compared in the Figure 4.



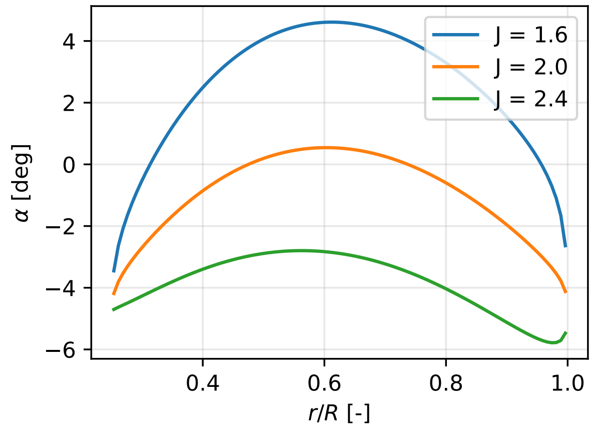
(a) Axial induction factor a.



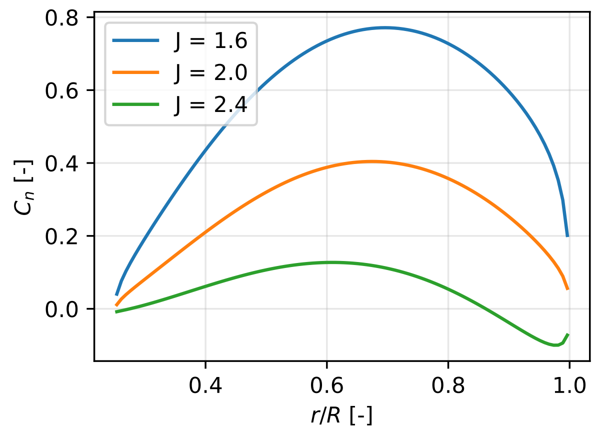
(b) Tangential induction factor a'.



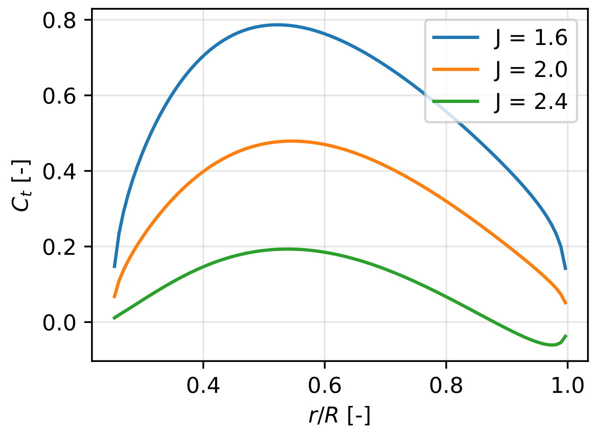
(c) Inflow angle ϕ.



(d) Angle of attack α.



(e) Normal force coefficient C_n.



(f) Tangential force coefficient C_t.

Figure 4: Spanwise distributions of induction factors (top), flow angles (middle), and force coefficients (bottom) for advance ratios $J = 1.6$, 2.0, and 2.4.

General Trends

There is a common trend that can be observed in all graphs. The advance ratio is a a proxy for the local angle of attack at each blade: as the advance ratio is reduced the angle of attack increases. This increases the loading, which increases the induction. The increasing induction decreases the inflow angle.

Another interesting relationship that can be seen in all plots is the negative loading at high advance ratios. Once the blade section starts operating below the zero lift angle of attack, the induction factor becomes negative, meaning the blade section starts extracting energy from the flow. From the ARAD8% polar, this is around $-4.5°$.

Induction Factors

The axial and tangential induction factors can be seen in figs. 4a and 4b. It can be seen that near the tip of the blade, the magnitude of the induction factor increases due to the root and tip effects. Overall, for most of the blade span, the axial induction factor increases towards the tip, whilst the tangential induction factor decreases towards the tip. This is due to the fact that the inflow angle is the highest at the root, which will result in a higher a'.

Flow Angles

The spanwise distribution ϕ angles can be seen in Figure 4c. For all advanced ratios, it can be seen that the inflow angle is the highest at the root of the propeller and decreases along the span. This is expected as the radial coordinate at the root is lower, the azimuthal velocity will be lower. With an increase in the advance ratio, the inflow angle increases as the axial velocity becomes more dominant compared to the tangential velocity. The behavior of ϕ at the tip of the rotor can follow the same trend as a'.

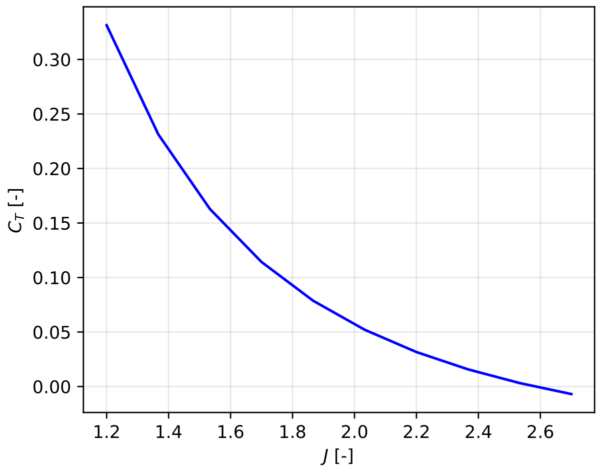
Looking at Figure 4d, it can be seen that the angle of attack decreases towards the root and tip and is the highest at around 0.6 of the span. The distribution of alpha is determined by the distribution of ϕ and the geometric twist distribution of the propeller. The twist distribution follows a linear relationship given in Table 1 and is kept the same for all cases. Therefore, to change the α of the propeller, the distribution of β and ϕ both need to be considered to get the wanted values of α.

Force Coefficients

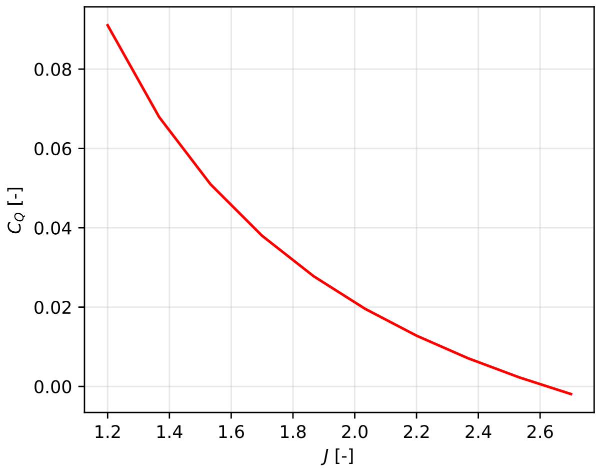
The spanwise distribution of the normal force coefficient can be seen in Figure 4e. The highest normal force coefficient can be found around $(r/R) = 0.7$ for $J = 1.6$, but it can be seen that the peak shifts towards the root with increasing J and reduces in magnitude. Additionally, the force on the tip is reduced to the tip effect and tip vortices, which reduces the loading on the blade section. A similar effect happens at the root. This normal force coefficient is more closely related to the trends in axial induction.

The spanwise distribution of the tangential force can be seen in Figure 4f. It exhibits a different distribution than C_n as the peak can be found more towards the root, but it also decreases with J. C_t is proportional to the product of relative velocity and a', therefore, it is higher closer to the root. The tip and root effects also impact the C_t, therefore, it tends to 0 at the root and tip. With increasing J, the relative velocity tends to decrease, and the a' also decreases; therefore, it is expected that C_t will also decrease. The phenomenon of why the peak shifts outboard can be explained by the fact that the inflow angle ϕ difference is greater at the tip with increasing J. Due to the higher increase in ϕ angles at the tip with respect to the increase of ϕ at the root, the total C_t will shift outboard as the force is proportional to $C_l \sin(\phi)$.

5.3 Thrust and Torque Coefficients versus Advance Ratio



(a) Thrust Coefficient C_T



(b) Torque Coefficient C_Q

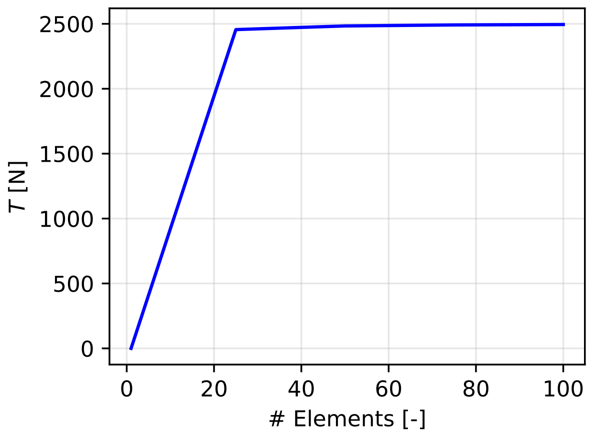
Figure 5: Thrust and Torque versus Advance Ratio

The thrust and torque coefficient have an inverse relation with the advance ratio, as can be seen in Figure 5. Since J is a proxy of the local angle of attack, as the advance ratio is reduced, the angle of attack at every section is increased, which increases the loading. This is reflected with an increasing C_T and C_Q for as J is decreased.

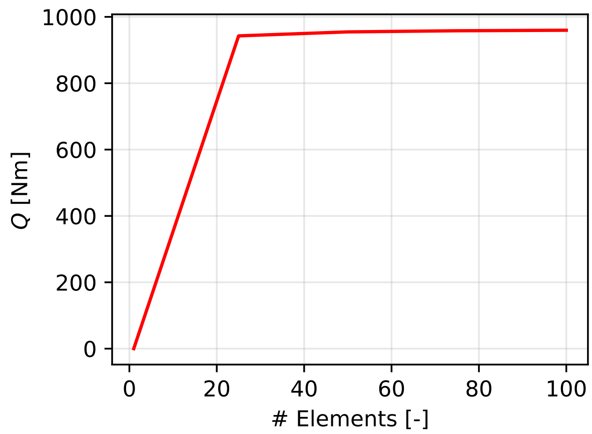
5.4 Convergence and Discretization

In this subsection the influence of number of annuli, spacing method and convergence history of total thrust is evaluated.

The influence of number of annuli is analyzed by plotting the thrust and torque versus the number of elements in Figure 6. With only 30 elements it can be seen that the thrust and torque have converged to their final values. The discretization error becomes insignificant with higher mesh resolution.



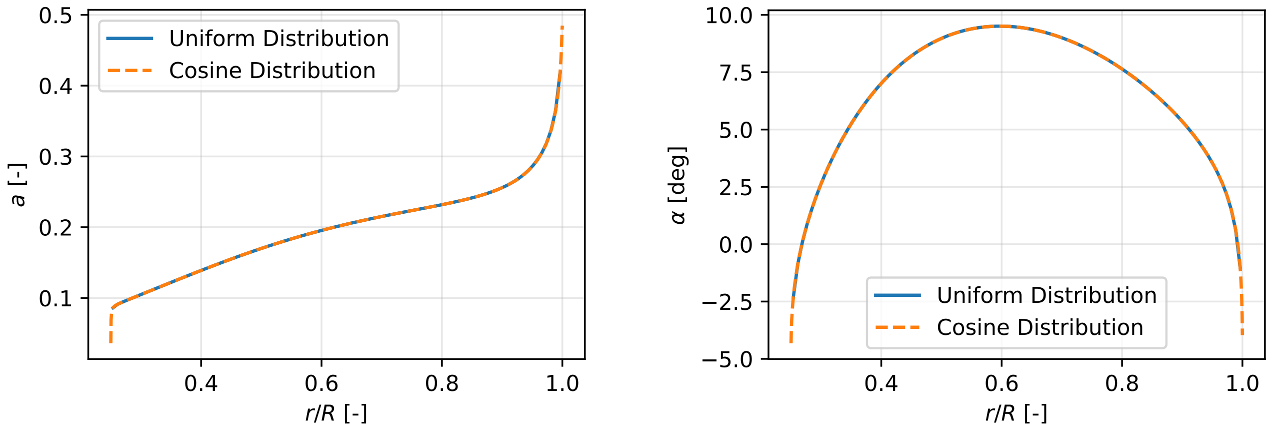
(a) Thrust



(b) Torque

Figure 6: Thrust and Torque versus Number of Annuli

The spacing method is evaluated by comparing the radial distribution of a and α in Figure 7, and by comparing thrust convergence history in Figure 8. A sufficient number of elements (100) is chosen to ensure convergence of the discretization, following Figure 6. The a and α graphs for the uniform and cosine distribution collapse on each other everywhere except at the edges where the cosine distribution appear more resolved. This is as expected; the cosine distribution places more elements near the hub and the tip. Since the blade edges experience higher velocity gradients, the cosine spacing is more suitable for capturing these effects.



(a) Axial Induction Factor a

(b) Angle of Attack α

Figure 7: Comparing Radial Distributions

The convergence history of the total thrust for both spacing methods show similar convergence. There is a slight thrust discrepancy between the two methods, which shows the significance of sufficient edge refinement.

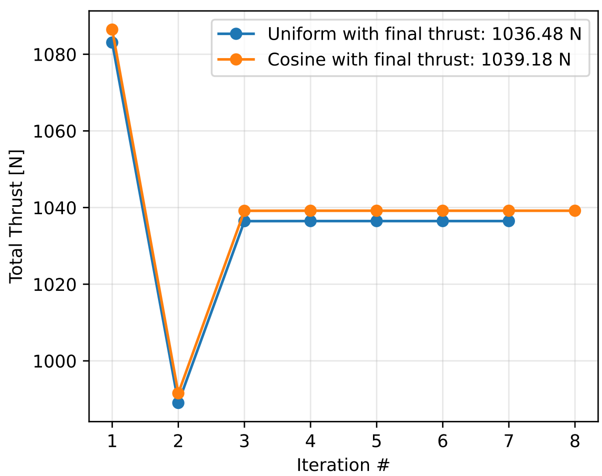


Figure 8: Thrust Convergence History

Fewer than 10 iterations are required for convergence, owing to the efficient root-solving bracketing scheme.

6 Propeller Geometry Optimization for Energy Harvesting

A benefit of propeller rotors mounted on electric engines is that they can be made to produce power and act as a wind turbine. This is an interesting development for electric-power aircraft under development, that could harvest energy in approach and landing phases.

We are tasked to modify the propeller blade geometry to maximize the power coefficient in this regime, for the given operating specifications Table 2. To achieve end, an optimization was implemented as formulated in eq. (4). For simplicity, the linear distribution of the twist and chord distributions was maintained as shown in Equation 5, resulting in only 4 design variables to be optimized (a_t, b_t, a_c, b_c). The collective pitch was included in the b_t variable to a well-posed problem. A constraint was added to ensure that the angle of attack at each annuli remain within that of the provided data. This, although limits the design space, ensures a physical solution.

$$
\begin{array}{lll}
\text{minimize} & C_P(\mathbf{x}) & \\
\text{by varying} & \underline{x}_i \leq x_i \leq \overline{x}_i & i = 1, \ldots, 4 \\
\text{subject to} & \alpha(\mathbf{x}) - \alpha_{\max} \leq 0 & \\
 & \alpha_{\min} - \alpha(\mathbf{x}) \leq 0 &
\end{array}
\tag{4}
$$

Representative bounds that restrict the design space were chosen following table 3.

$$
\begin{aligned}
\beta\left(\frac{r}{R}\right) &= a_t \cdot \frac{r}{R} + b_t \\
\frac{c}{R}\left(\frac{r}{R}\right) &= a_c \cdot \frac{r}{R} + b_c
\end{aligned}
\tag{5}
$$

Table 3: Design variables and their bounds.

Variable	Lower bound	Upper bound
a_t	$-90°$	$90°$
b_t	10	120
a_c	-0.10	-0.02
b_c	0.10	0.30

Since it was found that the optimizer had troubles stepping far away from the original design, as the performance is very sensitive to the chord and twist distribution, it was chosen to sample the design space at 1017 points of which 1000 are random[1] and 17 are all permutations of the bounds. Only the ten best designs were used as individual starting points for the optimizer. The scipy.minimize optimizer using an SLSQP method was used with a Jacobian step size of 0.02 and a precision target of 1e-10.

The optimal chord and twist distribution are plotted Figure 9, with the values $a_t^* = -43.63$, $b_t^* = 60.87$, $a_c^* = -0.0305$ and $b_c^* = 0.2455$. Important distributions are plotted in fig. 10

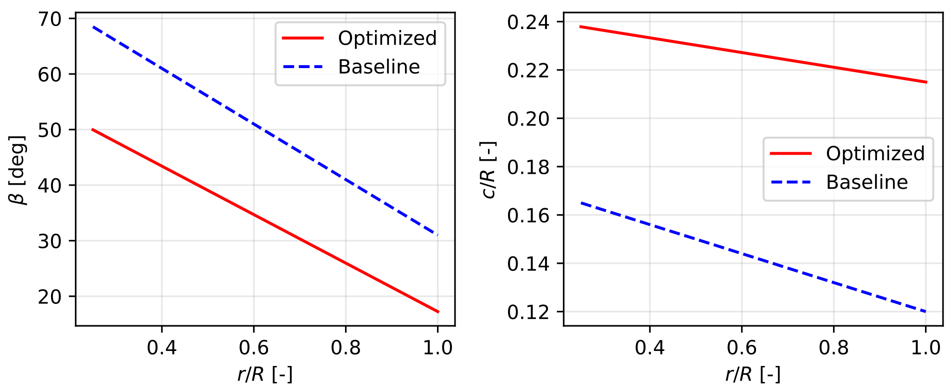


Figure 9: Geometric distribution of baseline and optimized propeller.

[1]To ensure reproducibility, a seed for the random scalar generation was chosen

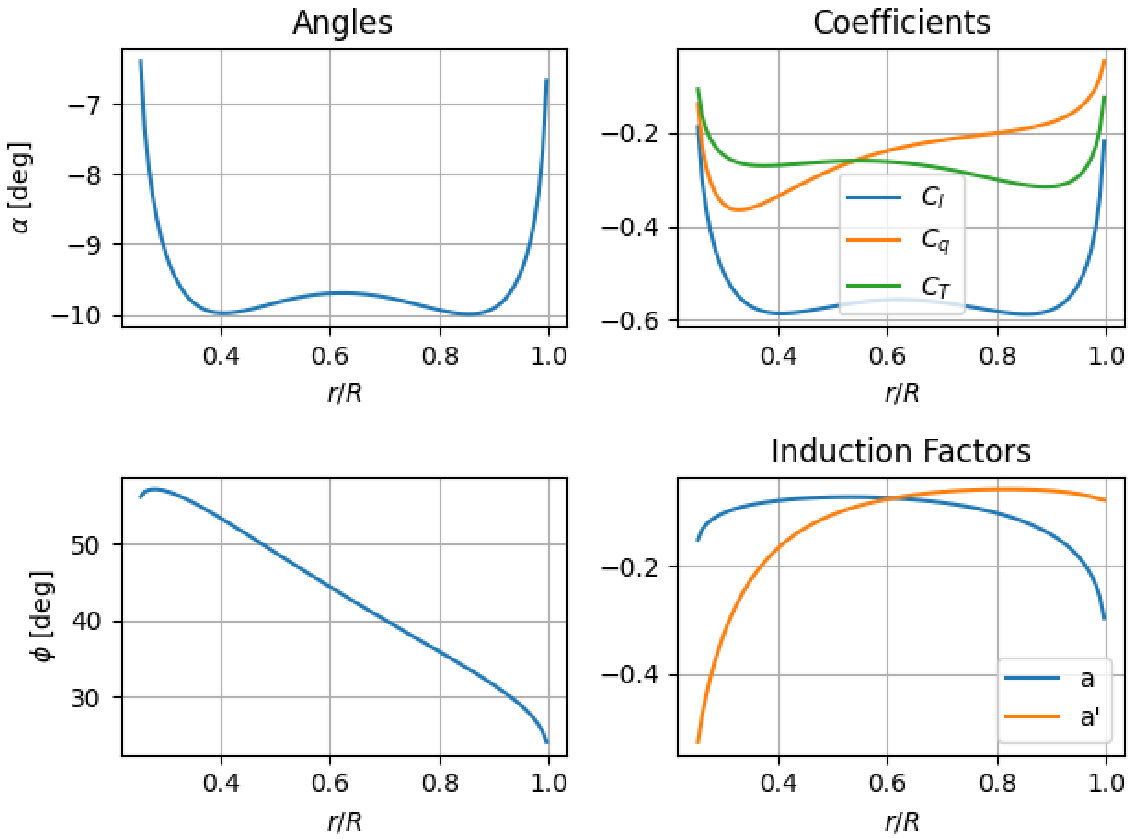


Figure 10: Spanwise distribution of key parameters of interest for the energy harvesting design.

The generated power for this design is equal to 95.429 KW with a $C_P = -0.5377$ [2]. From the actuator disk theory, the theoretical Betz limit $C_P = -16/27 = -0.593$ is associated with a power of $P = 105.12kW$ for generating propeller size and flow velocity. Comparing the two, the value is reasonable, as it is below the theoretical limit. This means that this design extracts 53.77% of the kinetic energy from the wind to spin the propeller compared to the Betz limit of 59.3 %. While this value is quite high, as common wind turbines extract 35% to 45% of the available kinetic energy to spin the propeller, this simulation is done under certain idealistic assumptions listed in section 4 and is optimized for one specific operating condition [5], so in reality, we expect a lower power harvesting capacity.

The twist distributions were optimized in a way such that the airfoil produces the greatest torque to be able to harvest the most power from the flow. This was expected to be close to values of $\alpha = -9.5$, as that is the point of maximum C_l/C_d. The optimizer therefore saw benefits in adjusting the twist such that the airfoil operates at a α close to the most negative C_l/C_d for most of the blade sections.

The chord distribution was changed in a way such that the overall chords of all blade sections are increased, and there is a smaller decrease along the span. This was expected as this will result in more torque being produced, which will maximize the harvested power. In the real world, there are weight penalties of having a large chord at the tip, but the structural calculations were not included in the optimizer, which is why it didn't penalize such a distribution.

The original (baseline) design requires 33.157 KW of power under the same operating conditions, meaning it wouldn't generate any power in the harvesting state. On the other hand, the optimized design would not function during cruise operations. Since switching propeller geometry is not

[2]The power coefficient is negative as the propeller nomenclature is used in this report.

possible mid-flight, the rotor needs to be designed such that it can function efficiently in both operating conditions, or have the capability of variable pitch to be optimized in both operations.

7 Stagnation Pressure Analysis

The stagnation pressure of the flow was calculated and compared at four different locations: 1) infinity upwind of the rotor, 2) upwind side of the rotor, 3) downwind side of the rotor, and 4) infinity downwind of the rotor. Since the flow is assumed to be incompressible, Bernoulli's equation can be used to calculate how the static pressure changes with the magnitude of total velocity, which includes the axial and tangential velocities. Furthermore, the stagnation pressure can be calculated by adding the dynamic pressure to the static pressure, once again using the magnitude of velocity. At the rotor, just before and after it, the axial and tangential velocities are given by Equation 6 while in the far wake by Equation 7.

$$\begin{aligned} V_{\text{ax}} &= V_\infty(1+a) \\ V_{\text{tgt}} &= \Omega(r/R)\cdot R\cdot(1-a') \end{aligned} \tag{6}$$

$$\begin{aligned} V_{\text{ax}} &= V_\infty(1+2a) \\ V_{\text{tgt}} &= 2\Omega(r/R)\cdot R\cdot(1-a') \end{aligned} \tag{7}$$

The pressure jump across the rotor is determined by calculating the local thrust on each element ($C'_T q_\infty\, dA$) and dividing by the local area by Equation 8.

$$\Delta p = \frac{C'_T\, q_\infty\, dA}{dA} \tag{8}$$

Where q_∞ is free stream dynamic pressure (since the thrust coefficient is normalized by q_∞). The obtained results for the stagnation pressure at the 4 locations can be seen in Figure 11. These results were obtained by using the BEM results for $J = 2$, and at an ISA altitude of 2000 m. The stagnation pressure was normalized by the free stream stagnation pressure $p_{t,\infty} = 81308Pa$ found far upstream of the rotor.

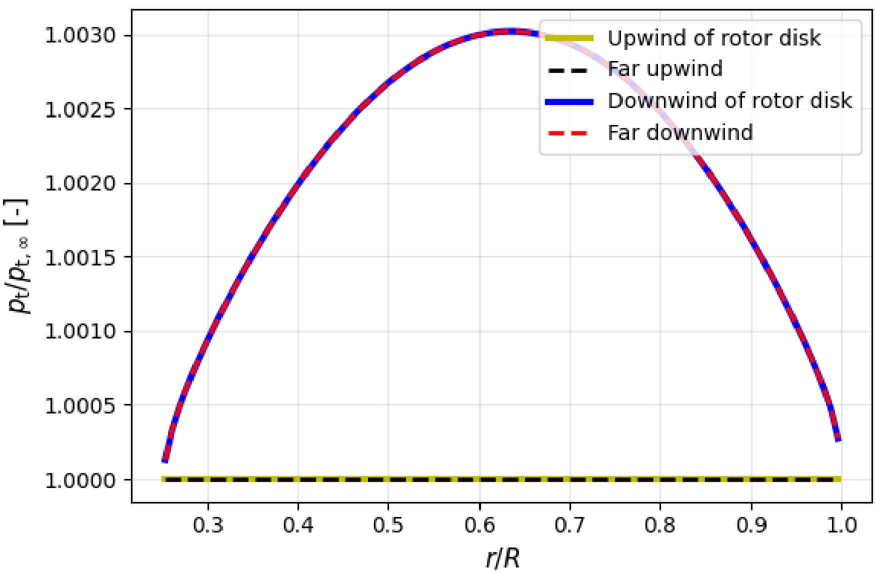


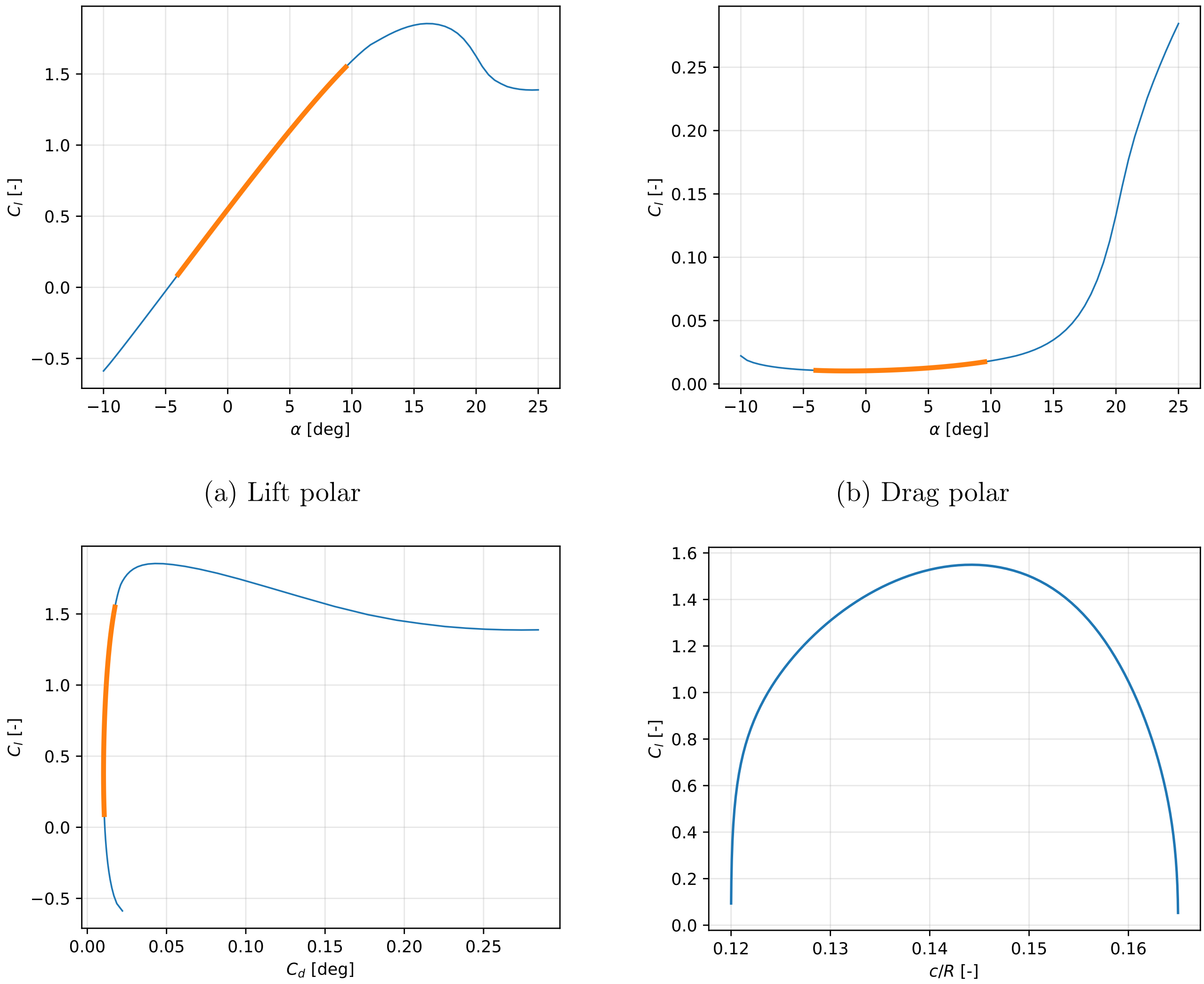
Figure 11: Stagnation pressure at different locations

It can be seen that far upwind and just upwind of the rotor, the stagnation pressure is the same, which is expected as there are no losses nor heat transfer mechanisms upstream. This is consistent with the assumptions in section 4. The same can be seen for the downwind of the rotor disk and far downwind locations. There is a pressure jump just before and just after the rotor due to the rotor itself that, being a propeller, introduces kinetic energy into the flow. This profile is what is expected and can be seen from other simulations [6].

8 Operational Point Lift and Drag

The blades airfoil cross-sections are analyzed in terms of their lift and drag coefficients. Polar plots displaying the utilized range of angle of attack with an orange line, provide a performance indication of the rotor in fig. 12. It can be seen that the operational range shown in orange is well within the drag bucket region, which is beneficial in the aerodynamic performance.

The lift coefficient can also be related to the normalized chord length c/R. Since the chord is a proxy for the spanwise position, fig. 12d is essentially a lift coefficient versus radial position plot. As expected, the lift near the hub and tip approach zero, while more lift is generated in the center of the blade.



(a) Lift polar

(b) Drag polar

(c) Lift-to-drag characteristics

(d) Lift coefficient versus normalized chord

Figure 12: Airfoil polars indicating the utilized range and the relation between lift coefficient and chord distribution.

9 Conclusion and Discussion

The BEM model has been developed to analyze a propeller rotor in a wide variety of operating conditions, as well as to optimize a propeller for energy harvesting. The robustness of the implementation has been demonstrated, not only through the various applications but also via the thorough verification against well established codes.

The results show that across these operating conditions, the propeller performance follows the expected trends. The importance of suitable discretization and iterating properties such as the number of elements, spacing distribution, and model selection has been underlined by evaluating their effects. The stagnation pressure in the radial direction is analyzed at four different positions, demonstrating the mechanical energy increase caused by the propeller. Another indicator of the rotor's performance is evaluating the lift and drag coefficients at the given operation. This proved that the propeller operates within a low drag region and, at a glance, shows good performance.

The assumptions have a significant impact on the results. However, for operating conditions within a reasonable range, the BEM model provides a representative prediction of the rotor performance. Further improvements can be made by considering the polar corrections for 3D flows and better extrapolation schemes. Alternatively, propeller-in-yaw condition could be considered.

References

[1] A. Ning, "Using blade element momentum methods with gradient-based design optimization," *Structural and Multidisciplinary Optimization*, 2021. [Online]. Available: `https://flowlab.groups.et.byu.net/preprints/Ning2020.pdf`.

[2] H. Glauert, "Airplane propellers," in *Aerodynamic Theory*, Springer, 1935, pp. 169–360.

[3] Airfoil Tools, *Ara-d 6% airfoil (arad6-il)*, `http://airfoiltools.com/airfoil/details?airfoil=arad6-il`, Accessed: 2026-03-23.

[4] T. Burton, N. Jenkins, D. Sharpe, and E. Bossanyi, *Wind Energy Handbook*, 2nd. Chichester, UK: John Wiley & Sons, 2011.

[5] Energy Education, *Betz limit*, `https://energyeducation.ca/encyclopedia/Betz_limit`, Accessed: 2026-03-23.

[6] R. E. Davidson, "Propeller lift and thrust distribution from wake surveys of stagnation conditions," NASA, Tech. Rep., 1952.